

Juniorprogrammierer.de

Go 1: Some new stuff

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Some more Go facts

- Today we will check out OOP in Go

- Please check out this link:

<https://www.youtube.com/watch?v=W3ZWbhQF6wg&list=PLoILbKo9rG3skRCj37Kn5Zj803hhiuRK6&index=18>

Method

- A method is a special type of function
- In go it has typically a receiver parameter before the function name
- Important:
 - No pointer => reading methods
 - Pointer => changing methods
 - For consistency if one method uses Pointer do it for all

```
type Person struct {  
    Name string  
    age int  
}  
  
func (p *Person) SetAge(age int) {  
    p.age = age  
}  
  
func main() {  
    person := Person{  
        Name: "Sascha",  
        age: 30,  
    }  
  
    person.SetAge(40)  
}
```

Setting attributes and methods private and public

- For attributes and for methods you use a small starting letter for private and a big starting letter for public
- This declares if other packages have access or not

```
type Person struct {  
    Name string  
    age int  
}  
  
func (p *Person) GetAge() int {  
    return p.age  
}  
  
func (p *Person) setAge(age int) {  
    p.age = age  
}
```

Interface

- An Interface specifies a required behaviour as method set
- Any type that implements that method set satisfies the interface

```
// Interface definieren
type Greeter interface {
    Greet() string
}
// Typ 1: Person
type Cat struct {
    Name string
}
func (p Cat) Greet() string {
    return "Hallo, ich bin " + p.Name
}
func SayHello(g Greeter) {
    fmt.Println(g.Greet())
}
func main() {
    cat := Cat{Name: "Sascha"}
    dog := Dog{Name: "Bello"}
    // Beide erfüllen das Greeter Interface
    SayHello(cat) // Hallo, ich bin Sascha
    SayHello(dog) // Wuff! Ich bin Bello
}
```

Exercise

1. Interface BankAccount with the following methods:
 - GetBalance() float64 - returns the account balance
 - Deposit(amount float64) - deposit money
 - Withdraw(amount float64) bool - withdraw money (return true if successful)
 - GetAccountType() string - returns the account type
2. Two Structs:
 - CheckingAccount - can go negative (overdraft up to -500)
 - SavingsAccount - cannot go negative
3. Function PrintAccountInfo(account BankAccount) that:
 - Prints the account type
 - Prints the current balance

Example Output:

Account Type: Checking Account
Balance: 1000.00

Account Type: Savings Account
Balance: 500.00